

Inorganic arsenic in rice bran and its products are an order of magnitude higher than in bulk grain.

Rice is more elevated in arsenic than all other grain crops tested to date, with whole grain (brown) rice having higher arsenic levels than polished (white). It is reported here that rice bran, both commercially purchased and specifically milled for this study, have levels of inorganic arsenic, a nonthreshold, class 1 carcinogen, reaching concentrations of approximately 1 mg/kg dry weight, around 10-20 fold higher than concentrations found in bulk grain. Although pure rice bran is used as a health food supplement, perhaps of more concern is rice bran solubles, which are marketed as a superfood and as a supplement to malnourished children in international aid programs. Five rice bran solubles products were tested, sourced from the United States and Japan, and were found to have 0.61-1.9 mg/kg inorganic arsenic. Manufactures recommend approximately 20 g servings of the rice bran solubles per day, which equates to a 0.012-0.038 mg intake of inorganic arsenic. There are no maximum concentration levels (MCLs) set for arsenic or its species in food stuffs. EU and U.S. water regulations, set at 0.01 mg/L total or inorganic arsenic, respectively, are based on the assumption that 1 L of water per day is consumed, i.e., 0.01 mg of arsenic/ day. At the manufacturers recommended rice bran solubles consumption rate, inorganic arsenic intake exceeds 0.01 mg/ day, remembering that rice bran solubles are targeted at malnourished children and that actual risk is based on mg kg⁽⁻¹⁾ day⁽⁻¹⁾ intake.